Why the zone-boundary mode is soft

A microscopic account of the L-point excitation in the pyrochlore ring-exchange model

This note explains, from the bottom up, why one excitation at the zone-boundary L point costs half of what its partner at the same momentum costs. The whole argument rests on one identity: in the ice basis the neutron probe is a diagonal *operator, so a ring exchange either changes the value it reads or it does not. Everything else follows.*

1 The one thing to remember about the energy

The model is the pure ring-exchange Hamiltonian on the spin-ice manifold,

$$H = -K \sum_p \left(W_p + W_p^\dagger\right), \qquad W_p = S_1^+ S_2^- S_3^+ S_4^- S_5^+ S_6^-, \tag{1}$$

where p runs over the elementary hexagons of the pyrochlore lattice. W_p takes an ice configuration in which the six spins around p alternate up–down–up–down–up–down and flips all six at once.

Equation (1) has no diagonal part. There is no potential energy, no field term, nothing but resonance. The immediate consequence is the only fact about energy we will need:

Energy is ring coherence, and nothing else. For any eigenstate $|n\rangle$,

$$\omega_n \;=\; E_n - E_0 \;=\; K \sum_p \Big[\underbrace{\langle 0|W_p + W_p^\dagger|0\rangle}_{\text{ground state}} - \underbrace{\langle n|W_p + W_p^\dagger|n\rangle}_{\text{excited state}}\Big] \;\equiv\; K \sum_p \Delta_p(n).$$

An excitation is a *bill*, itemised plaquette by plaquette. To cost little, a state must leave the resonance $\langle W_p + W_p^\dagger\rangle$ nearly intact on nearly every hexagon.

This is exact and additive, and at 48 sites we can read off every line of the bill from the exact eigenvector. That is what makes the question "why is this state cheap?" answerable rather than rhetorical.

2 In the ice basis the probe is diagonal

The emergent electric field is the spin itself, $E_\ell = S_\ell^z$, so the operator that neutrons couple to is

$$A_{\boldsymbol{q}}^\mu = \sum_{i\in\mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S_i^z, \tag{2}$$

one operator for each of the four pyrochlore sublattices μ. (The response S^{zz} sums $|\langle n|A_{\boldsymbol{q}}^\mu|0\rangle|^2$ over μ; the sublattice label matters, and we will see in §7 what goes wrong if it is dropped.)

Now the key structural point. We work in the basis of ice configurations $|c\rangle$. In that basis S_i^z is diagonal, so $A_{\boldsymbol{q}}^{\mu}$ is *diagonal too*:

$$A_{\boldsymbol{q}}^{\mu}|c\rangle = a^{\mu}(c)\,|c\rangle, \qquad a^{\mu}(c) = \sum_{i\in\mu} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_i} S_i^z(c). \tag{3}$$

Each configuration carries a single complex number $a^{\mu}(c)$: the amplitude of the imposed density wave in that configuration.

The ring exchange, by contrast, is purely off-diagonal: it takes $|c\rangle$ to $|c'\rangle$ with hexagon p flipped. So we can ask a completely concrete question:

when a ring exchange moves the system from c to c′, does the number the probe reads change?

Because only six spins move, the answer is a short computation, and it is the commutator $[A_{\boldsymbol{q}}^{\mu}, W_p] = g_p^{\mu}(\boldsymbol{q})W_p$ written as a statement about configurations:

$$\boxed{a^{\mu}(c') - a^{\mu}(c) = \pm\, g_p^{\mu}(\boldsymbol{q})} \qquad g_p^{\mu}(\boldsymbol{q}) = \sum_{j\in p\,\cap\,\mu} (-1)^j e^{-i\boldsymbol{q}\cdot\boldsymbol{r}_j}. \tag{4}$$

The quantity g_p^{μ} — the alternating lattice curl of the imposed wave around plaquette p — is *exactly the amount by which flipping p shifts the density-wave amplitude.* We verified Eq. (4) directly on 4000 ice configurations, every hexagon, every sublattice: the largest discrepancy was 0.000×10^0.

3 Geometry: some plaquettes are blind to the wave

Equation (4) turns a dynamical question into a geometric one. When is $g_p^{\mu} = 0$?

A pyrochlore hexagon is planar, and its plane is perpendicular to one of the four $\langle 111\rangle$ axes. Each hexagon omits exactly one sublattice, and that omitted label μ names its axis, so the hexagons fall into four *families* of equal size (twelve each on our 48-site cluster).

A plane wave $e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}$ is constant on planes perpendicular to $\boldsymbol{q}$ — its wavefronts. So if a hexagon's plane *is* a wavefront, all six of its sites carry the same phase, the alternating sum cancels term by term, and $g_p^{\mu} = 0$ for every μ. That happens precisely when $\boldsymbol{q}$ is parallel to that family's $\langle 111\rangle$ axis — which is the case at L, where $\boldsymbol{q} \parallel [1\bar{1}1]$.
So at L one quarter of all ring terms are *blind* to the imposed wave: flipping them changes the number the probe reads by exactly nothing. We call that family *dark*, and the other three *bright*. This is a statement about the momentum, not about any state — a point we return to in §5.

4 Dephasing: why bright plaquettes cost energy

Now put the two ingredients together. Think of the excitation, to a first approximation, as the ground state *modulated* by the density wave,

$$\psi(c) \;\simeq\; a(c)\,\psi_0(c). \tag{5}$$

This is the single-mode (Feynman) state $A_{\boldsymbol{q}}|0\rangle$ written in components. Ask what its ring coherence is on plaquette p. Coherence is a sum over the pairs (c, c') that W_p connects:

$$\langle\psi|W_p|\psi\rangle \;\sim\; \sum_{c\,\to\,c'} a^*(c')\,a(c)\,\psi_0(c')\,\psi_0(c). \tag{6}$$

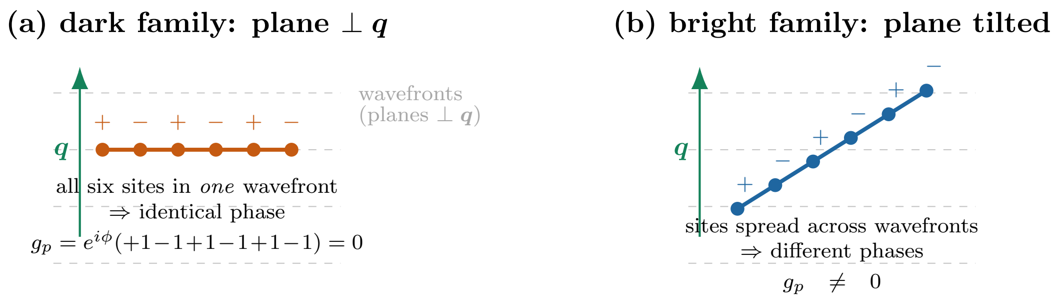

Figure 1: The geometric origin of a dark family, drawn edge-on so that wavefronts are lines. **(a)** When $\boldsymbol{q}$ is parallel to a family's $\langle 111 \rangle$ axis, that family's hexagons lie inside a single wavefront. All six sites carry the same phase, the alternating signs cancel in pairs, and g_p vanishes identically. **(b)** Every other family is tilted with respect to the wavefronts, its sites pick up different phases, and $g_p \neq 0$. At L the measured geometric weights $\sum_\mu |g_p^\mu|^2$ per family are $(96, 96, \mathbf{0}, 96)$; the dark family is exactly the one whose axis is parallel to $\boldsymbol{q}$, with $|\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{q}}| = 1.000000$.

The ground state is stoquastic, so $\psi_0 > 0$ and every term in the ground-state version of Eq. (6) adds constructively: that positive interference *is* the ground-state energy. In the excited state each term is multiplied by $a^*(c')a(c)$, and by Eq. (4) that factor depends entirely on whether p is dark or bright.

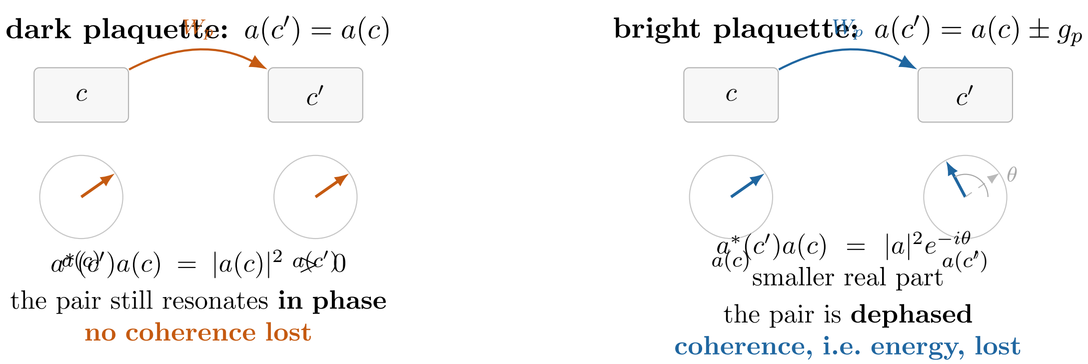

Figure 2: The mechanism in one picture. A ring exchange connects two ice configurations c and c'. Each carries a phasor $a(c)$, the local amplitude of the imposed density wave. On a *dark* plaquette the flip does not move the phasor at all, so the two configurations still interfere constructively and the resonance survives untouched. On a *bright* plaquette the flip rotates the phasor by an amount set by g_p, the interference is spoiled, and the lost coherence appears as excitation energy.

This is also, finally, what the f-sum rule has been saying all along. The exact first moment of the response is

$$f_1(\boldsymbol{q}) = \frac{K}{2N} \sum_p \langle W_p + W_p^\dagger \rangle \sum_\mu \left| g_p^\mu(\boldsymbol{q}) \right|^2, \tag{7}$$

a formula that had always looked like a geometric weight multiplying a measured coherence. Equation (4) says what the weight *is*: $|g_p^\mu|^2$ is the dephasing that the imposed wave inflicts on the ring resonance at p. The oscillator strength is the total dephasing, and it is blind to the dark family because a dark plaquette cannot be dephased.

5 The bill, and where the relaxation happens

Everything so far concerns the *momentum*: at L a dark family exists, and the rigid wave (5) pays only on bright plaquettes. That is not yet an explanation of why one eigenstate is soft and its partner is not, because the dark family is equally available to both. The distinction is what each state *does* with it.

The rigid wave is a poor state and the true eigenstates relax away from it. Reading the itemised bill off the exact eigenvectors — projected onto the state the probe actually creates at $\boldsymbol{q}_L$, so that the answer is basis-independent — gives Table 1 and Fig. 3.

Table 1: Excitation energy in units of K at L, 48 sites, split by whether the probe drives the family.

state	3 bright families	1 dark family	total
rigid wave $A_{\boldsymbol{q}}\vert 0\rangle$	0.968	0.878	1.846
soft eigenstate	**0.108**	0.915	**1.024**
partner	1.082	0.648	1.730
relaxation (rigid $\to$ soft)	0.860 (105%)	-0.038 (-5%)	0.822

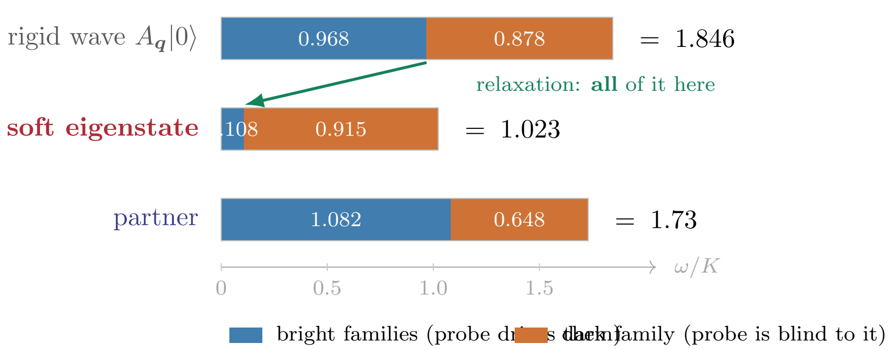

Figure 3: The itemised energy bill at L. The rigid density wave pays $0.968K$ of dephasing on the three bright families. The true soft eigenstate has recovered almost all of it — $0.968 \to 0.108$, an 89% restoration — while its dark-family cost is essentially unchanged. *The entire relaxation happens on the bright families.* The partner, living in the orthogonal transverse channel, does not perform this recovery and pays ten times more where the probe drives.

Read Fig. 3 from the top down and the answer to “why is this state cheap?” is immediate:

> The soft mode is a density wave **dressed so as to re-phase the ring resonances the wave would otherwise dephase**. It restores 89% of the coherence on the thirty-six plaquettes the probe drives, and it can afford to leave its residual disturbance on the twelve dark ones, because flipping those does not change the density-wave amplitude [Eq. (4)] and therefore costs it no spectral weight. It buys cheap energy with a currency the probe cannot see.

That last clause is what makes the trade possible rather than merely desirable. A state could always lower its energy by abandoning the density wave — but then it would no longer be created by the probe and would carry no weight in S^{zz}. The dark family is the one place where a state can rearrange itself *without* paying that price.

6 Why only at L, and why only one of the two levels

Two controls make the argument tight.

Momentum control (X). At X no $\langle 111\rangle$ axis is parallel to $\boldsymbol{q}$, so no family is dark: the geometric weights are $(96, 96, 96, 96)$. There is nowhere to park the disturbance, the trade of Fig. 3 is unavailable, and nothing softens — the lowest X level sits at 0.952 of its Gaussian energy, against 0.503 at L. Softening tracks the *existence* of a dark family.

State control (the partner). At L the dark family is available to every level, yet only one exploits it. The soft mode pays $0.108K$ on bright, the partner $1.082K$ — a factor of ten, against an overall energy ratio of only 0.59. Availability is a property of the momentum; exploitation is a property of the state. The same pattern holds at 72 sites, where the bright-family ratio is 0.282 against an overall ratio of 0.589.

7 A caveat that cost us a wrong figure

The sublattice label in Eq. (2) is not decoration. The sublattice-*summed* operator $A_{\boldsymbol{q}} = \sum_\mu A^\mu_{\boldsymbol{q}}$ commutes with *every* ring exchange at L: its total alternating sum vanishes on every hexagon, dark and bright alike. A check built on that operator returns zero for both families and appears to confirm the mechanism while actually testing nothing.

The correct statement uses the resolved operators, and it is stronger than "the total cancels": on a dark plaquette *each* g^μ_p vanishes separately, so all four sublattice amplitudes are individually unmoved by the flip. The cross-check is that $\sum_\mu |g^\mu_p|^2$ summed over each family reproduces the geometric weights $(96, 96, 0, 96)$ exactly.

8 The computed evidence

Everything above was asserted; this section is the arithmetic behind it. All numbers come from exact diagonalization of the ground winding sector of the 48-site cluster fcc$(2, 2, 3)$ (dimension 9 104, $E_0 = -10.55601176\,K$) unless another size is named, and every script that produced them is in the repository.

Gate. The family decomposition used throughout must reconstruct the Hamiltonian. It does: $\sum_\mu\langle 0|H_\mu|0\rangle = -10.5560117565$ against $E_0 = -10.5560117565$, a difference of 2.8×10^{-14}.

8.1 The flip identity, Eq. (4)

For 4000 ice configurations, every hexagon, every sublattice, we flipped each flippable plaquette and compared the change in the density-wave amplitude against g^μ_p. The identity is exact (Table 2); note in particular that the dark family does not merely have a small shift, it has none at all.

Table 2: Direct verification of $a^\mu(c') - a^\mu(c) = \pm g_p^\mu$ at $\boldsymbol{q}_L$. The sign is per configuration (which way the ring flips).

family μ	n_{hex}	$\max_\mu \lvert g_p^\mu \rvert$	$\max\lvert \Delta a^\mu \mp g_p^\mu \rvert$	mean $\lvert \Delta a^\mu \rvert$
0	12	2.000000	0.000×10^0	1.000000
1	12	2.000000	0.000×10^0	1.000000
2 (dark)	12	**0.000000**	0.000×10^0	**0.000000**
3	12	2.000000	0.000×10^0	1.000000

8.2 The dark family is the one whose axis is parallel to $\boldsymbol{q}$

Table 3 confirms the wavefront picture of Fig. 1 quantitatively: the family with vanishing geometric weight is precisely the family whose $\langle 111 \rangle$ axis is parallel to $\boldsymbol{q}_L$, to six digits.

Table 3: Geometric weight $\sum_p \sum_\mu |g_p^\mu|^2$ per family at $\boldsymbol{q}_L = (\frac{1}{2}, -\frac{1}{2}, \frac{1}{2})$, against the orientation of each family's axis.

family μ	n_{hex}	$\sum \lvert g \rvert^2$	$\langle 111 \rangle$ axis	$\lvert \hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{q}} \rvert$
0	12	96.0000	$[1\,1\,1]$	0.333333
1	12	96.0000	$[1\,\bar{1}\,\bar{1}]$	0.333333
2 (dark)	12	**0.0000**	$[\bar{1}\,1\,\bar{1}]$	**1.000000**
3	12	96.0000	$[\bar{1}\,\bar{1}\,1]$	0.333333
at $X = (1,0,0)$:		96.0, 96.0, 96.0, 96.0		— no family is dark

The ground state does resonate on the dark family, so the cancellation is not trivially about idle plaquettes: the family-resolved coherences are $\langle W_p + W_p^\dagger \rangle_\mu = 2.6296, 3.2195, 2.3535, 2.3535$, summing to $10.5560 = |E_0|$. (They are unequal because the $2 \times 2 \times 3$ box makes the four families inequivalent; this is a finite-size effect, not physics.)

8.3 The energy bill, and the two controls

Table 1 in §5 gives the 48-site ledger. Table 4 adds what makes it an argument rather than an observation.

Two things to read off. First, at both sizes the soft level's overall advantage (0.59) is far smaller than its advantage on the driven plaquettes (0.10 and 0.28): the saving is specifically there, not spread evenly. Second, at X the trade is simply unavailable, and the lowest level sits at 0.952 of its Gaussian energy instead of 0.503. Softening tracks the *existence* of a dark family.

8.4 The consequence: a non-monotonic branch

If the mechanism is right it should bend the dispersion, and it does. Walking outward along $\Gamma \to L$, the branch rises like a photon, turns over, and falls into the zone boundary (Fig. 4). Every cluster we can diagonalize shows it.

8.5 What the ground state already knew

Two ground-state quantities anticipate all of this, which is worth recording because they are computable far beyond exact diagonalization. The static structure factor is largest exactly where the

Table 4: Excitation energy in K, split by whether the probe drives the family. The soft level's advantage is concentrated entirely in the bright column; at X, where no family is dark, no such split exists and nothing softens.

	state	bright	dark	total	$\omega_{\text{low}}/\omega_{\text{Gauss}}$
48 sites, L	soft	0.1081	0.9154	1.0235	0.503
	partner	1.0822	0.6481	1.7303	0.851
	ratio	**0.0999**	1.4124	0.5915	
72 sites, L	soft	0.4585	0.6095	1.0681	
	partner	1.6237	0.1906	1.8143	
	ratio	**0.2824**	3.1973	0.5887	
48 sites, X	dominant	no dark family exists		2.2354	0.952

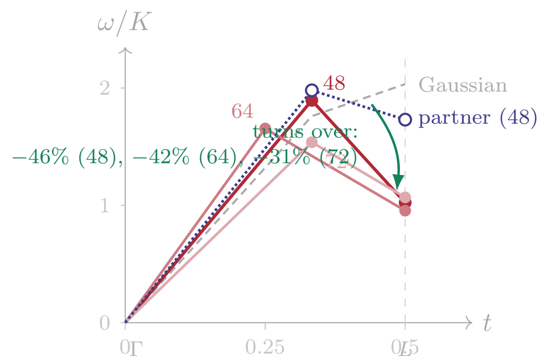

Figure 4: The lower branch along $\Gamma \to L$, with $\boldsymbol{q} = t(1, \bar{1}1)$. It rises like a photon — at $t = \frac{1}{3}$ it is *above* the Gaussian curve, and nearly degenerate with its partner (1.895 against 1.980) — then turns over and falls into the zone boundary. The partner falls only 12.6% over the same interval. Each cluster resolves one interior momentum on this line, so what is established is the inequality $\omega(\text{interior}) > \omega(L)$ at three independent sizes, not the shape of the minimum.

mode softens ($S(L) = 0.3704$, against 0.2872, 0.2729, 0.2583, 0.1894 at the other resolved momenta), and the Feynman ratio f_1/S, normalized by the Gaussian energy, orders all five momenta exactly as the measured softening does — Spearman correlation +1, no inversions.

8.6 The baseline: no Gaussian theory can do this

Finally, the reference the softening is measured against. In Gaussian lattice electrodynamics the two transverse polarizations are not merely close but *exactly* degenerate: diagonalizing the lattice curl at every allowed momentum of fcc$(4,4,4)$, fcc$(3,4,5)$ and fcc$(2,3,5)$ — 153 momenta in all, including 32 in the generic class on the anisotropic boxes — gives $\sigma_2/\sigma_1 = 1.00000000$ every time, with $\sigma_3/\sigma_1 \sim 10^{-15}$ confirming rank two.

Since U and K enter $\omega_\lambda = \sqrt{2UK}\,\sigma_\lambda$ only through the common prefactor, *no* renormalization of the Gaussian theory — of any strength, in either parameter — can split the doublet. The observed factor 1.69 at L is therefore outside that description entirely, not a large correction within it.

Table 5: The ground-state predictor against the measured softening. The two right-hand columns rank the five momenta identically. Neither is ordered by $|\boldsymbol{q}|$: the first and last rows lie on the same shell.

$\boldsymbol{q}$	$\lvert\boldsymbol{q}\rvert$	$S(\boldsymbol{q})$	f_1	$(f_1/S)/\omega_G$	$\omega_{\text{low}}/\omega_G$
L	0.866	0.3704	0.6835	**0.908**	**0.503**
$(\frac{1}{3},\frac{1}{3},\frac{2}{3})$	0.816	0.2870	0.7150	1.177	0.824
X	1.000	0.2727	0.8790	1.373	0.952
$(\frac{1}{3},\frac{1}{3},\frac{1}{3})$	0.577	0.1894	0.4585	1.375	1.076
$(\frac{1}{6},\frac{1}{6},\frac{5}{6})$	0.866	0.2583	0.8306	1.415	1.114

8.7 What this does not establish

Three things, stated so they are not read into the above.

This is not a new quasiparticle. The soft level is one of the two transverse polarizations, continuously connected to the photon doublet at small $\boldsymbol{q}$ (at $t = \frac{1}{3}$ the two levels are 1.895 and 1.980, nearly degenerate), carrying no new quantum number and no new charge. What the mechanism describes is the *same* excitation reorganizing itself where the lattice permits — which is exactly the status of the roton in helium, where phonon and roton are one branch.

The minimum is not resolved. Each cluster contributes one interior point on the $\Gamma \to L$ line, so we establish $\omega(\text{interior}) > \omega(L)$ three times over, not the curvature or position of a minimum.

The splitting itself is not new. That interactions split the two transverse polarizations at intermediate wavevector was predicted semiclassically by Kwasigroch, Douçot and Castelnovo. What is added here is that the splitting is order one at the zone boundary, that one member falls to half the Gaussian energy while the other does not, and the exact microscopic reason why.

9 Summary

1. Energy in this model *is* ring coherence; an excitation is an itemised bill over plaquettes.
2. In the ice basis the probe is diagonal, so a ring exchange either shifts the density-wave amplitude or it does not: $a^\mu(c') - a^\mu(c) = \pm g_p^\mu$, verified to 0.000×10^0.
3. Hexagons lying inside a wavefront have $g_p \equiv 0$: they are blind to the wave. At L that is exactly one of the four families, 25% of all ring terms.
4. Dephasing is the energy: a flip that moves the phasor spoils the constructive interference that made the ground state cheap. Hence $f_1 \propto \sum_p \langle W_p + W_p^\dagger\rangle |g_p|^2$ — the oscillator strength *is* the total dephasing.
5. The rigid wave pays $0.968K$ of dephasing on bright plaquettes. The true soft eigenstate recovers 89% of it and parks the remainder on the dark family, where rearrangement is free of spectral-weight cost. Its partner does not, and pays ten times more.
6. No dark family, no softening: X is the control.